

SAMUEL MICHAELS RODRIGUEZ

Machine Learning / AI Engineer

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PROFESSIONAL SUMMARY

Computer Science honours graduate (B.Sc., 3.9 GPA, Dean's List) and ML / software engineer who ships deployed systems end to end. Raised a research model's detection F1 from 0.42 to 0.77, built an AI phishing-detection browser extension on a FastAPI + RAG backend, and delivered a production React web app for a paying client. Fluent in Python, PyTorch, FastAPI, React, and Docker, with a track record of taking projects from raw data to deployment.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C, Bash, HTML/CSS

Machine Learning: PyTorch, HuggingFace Transformers, scikit-learn, XGBoost, NumPy, pandas — transformer fine-tuning, NLP / sequence labelling, CNNs, reinforcement learning (DQN, PPO), computer vision, statistical model evaluation

LLM / Generative AI: RAG pipelines, LangChain, ChromaDB & vector databases, embeddings, prompt engineering, LLM evaluation, agentic workflows, Ollama, Groq, Anthropic / OpenAI APIs

Backend & Web: FastAPI, REST API design, Next.js, React, Vite, Tailwind CSS, Node.js, Redis, PostgreSQL, SQLite, Streamlit

MLOps & Tooling: Docker, Git / GitHub, CI/CD (GitHub Actions), Linux, Railway, Render, HuggingFace Spaces, Jupyter, LaTeX, AI-assisted development (Claude Code, Cursor)

EXPERIENCE

Machine Learning Research Developer

May 2025 – Apr 2026

Western University — London, ON

- Increased syllable-boundary detection accuracy from an F1 of 0.42 to 0.77 — an 83% relative gain, statistically validated at $p = 0.016$ (Cohen's $d = 2.61$) — by augmenting the VG-HuBERT transformer with a task-specific segmentation head in PyTorch.
- Delivered a fully reproducible evaluation framework across 6 subjects by engineering a leave-one-run-out cross-validation pipeline in Python with time-aligned ground-truth labels at 30 ms tolerance.
- Kept experiments reliable and repeatable by running and version-controlling all model training and evaluation on Linux with Git, producing publication-quality analysis and figures.

Software Developer (Research)

May 2024 – Sept 2024

Western University — London, ON

- Eliminated 100% of detected memory leaks and removed extensive duplicated code from a legacy C simulation (bacterial-growth modelling) by refactoring monolithic logic into clean, modular functions on Linux/Unix.
- Improved the tool's runtime efficiency and long-term maintainability by restructuring the codebase and standardizing its layout, making it easier for the lab to extend.

Freelance Software Developer

2025 – 2026

Self-employed — London, ON

- Shipped a production, multi-page web application (SigmaK9 Training) for a local business by building it in React (Vite) with Tailwind CSS, client-side routing via React Router, and a filterable catalog of 12+ products across 4 categories, deployed to Vercel.
- Delivered the product on schedule and to spec by translating a non-technical client's requirements into working features through short, iterative feedback cycles — an agile, communication-driven workflow.

Produce Supervisor

Sep 2020 – Present

Sunripe Fresh Market — London, ON

- Led a 4–6 person crew and cut shelf-tag scanning errors to zero while completing a full-time CS degree, demonstrating reliability, ownership, and teamwork under pressure.

PROJECTS

PhishGuard — AI Phishing Detection (Browser Extension + API)

Python, FastAPI, Redis, LLM/RAG, ChromaDB, Docker

- Built a cross-browser WebExtension (Chrome & Firefox, Manifest V3) backed by a FastAPI service that scores URLs in real time and returns an explainable risk verdict.
- Engineered a multi-signal detection pipeline combining fast heuristics, typosquatting detection, WHOIS domain-age checks, VirusTotal reputation scoring, and an LLM + RAG layer (ChromaDB over threat-intelligence sources) for context-grounded reasoning.
- Added a versioned Redis cache to keep latency low and API costs down; debugged and fixed real production issues including stale cache verdicts and false-positive scoring. Deployed via Railway.

RAG Document Intelligence API

LangChain, ChromaDB, FastAPI, Docker, HuggingFace Spaces

- Built and deployed a retrieval-augmented document Q&A service: PDF ingestion, chunking, embedding, and semantic retrieval over a ChromaDB vector store, exposed through a documented FastAPI REST API.
- Grounded LLM responses in retrieved context to prevent hallucination; containerized with Docker (persistent vector-store volume) and covered with automated pytest tests before deploying to HuggingFace Spaces.

Connect Four Reinforcement Learning Agents

PyTorch — DQN, PPO, CNNs (academic paper)

- Trained and compared DQN and PPO agents for Connect Four and wrote a NeurIPS-format paper; novel contributions included supervised pretraining, a dual-path CNN, and a hybrid minimax–RL inference agent.

YOLOv8 Adversarial Robustness for Traffic-Sign Detection

PyTorch, YOLOv8, Albumentations (academic paper)

- Trained and evaluated YOLOv8 object-detection models on a 264-class traffic-sign dataset across adversarial conditions (tape, patch, graffiti, illumination), using synthetic augmentation to improve robustness.

EDUCATION

B.Sc. Honours, Computer Science

June 2026

Western University — London, ON

3.9 GPA • Dean's List • All coursework completed

Relevant coursework: Machine Learning, Deep Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Data Structures & Algorithms, Databases.